

UNIVERSITY OF BRISTOL

Faculty of Science

**MOCK EXAMINATION — SOLUTIONS AND
MARK SCHEME****Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012 — Paper 1

Total Marks: 100

Marking Notes:

- Mark allocations are shown in **blue** at each step.
 - Final answers are highlighted in green boxes.
 - “*(From formula sheet)*” indicates the equation is on the provided formula sheet.
 - “*(Recalled — not on formula sheet)*” indicates the equation must be known from memory.
 - Award partial marks for correct method even if the final answer is wrong.
 - Accept equivalent correct forms of answers.
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Section A: Oscillations and Waves — Answers

1. **B** — The displacement is $x(t) = 0.05 \cos(6\pi t + \pi/4)$. The angular frequency is $\omega = 6\pi$ rad/s. (*From formula sheet*) The period is $T = 2\pi/\omega = 2\pi/(6\pi) = 1/3$ s.

2. **B** — The total energy is $E = \frac{1}{2}kA^2$. (*Recalled — not on formula sheet*) At maximum speed all energy is kinetic: $\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kA^2$, so $v_{\max} = A\sqrt{k/m} = 0.08 \times \sqrt{160/0.4} = 0.08 \times 20 = 1.60$ m/s.

3. **B** — We have $\omega_0 = \sqrt{k/m} = \sqrt{100/0.25} = \sqrt{400} = 20$ rad/s and $\gamma = b/m = 10/0.25 = 40$ s⁻¹. (*Recalled — not on formula sheet*) The critical damping condition is $\gamma = 2\omega_0$. Here $2\omega_0 = 2 \times 20 = 40$ s⁻¹, and $\gamma = 40$ s⁻¹, so $\gamma = 2\omega_0$ exactly. The system is **critically damped**.

4. **A** — At (amplitude) resonance, the displacement amplitude is maximum and the phase lag δ of the displacement behind the driving force satisfies $\tan \delta \rightarrow \infty$, giving $\delta = \pi/2$. (*From formula sheet*) (from $\tan \delta = \gamma\omega/(\omega_0^2 - \omega^2)$, and at resonance $\omega \approx \omega_0$ for light damping, so $\delta = \pi/2$).

5. **C** — The wave speed on a string is $v = \sqrt{T/\mu}$. (*From formula sheet*) $v = \sqrt{100/0.004} = \sqrt{25000} = 158$ m/s (to 3 significant figures).

6. **C** — The reflection coefficient is $r = (Z_1 - Z_2)/(Z_1 + Z_2)$. (*From formula sheet*) For no reflection, $r = 0$, which requires $Z_1 = Z_2$ (impedance matching).

7. **C** — $\beta = 10 \log_{10}(I/I_0) = 10 \log_{10}(0.01/10^{-12}) = 10 \log_{10}(10^{10}) = 10 \times 10 = 100$ dB. (*From formula sheet*)

8. **D** — For a source approaching a stationary observer: $f' = f \cdot v/(v - v_s)$. (*From formula sheet*) $f' = 600 \times 340/(340 - 34) = 600 \times 340/306 = 600 \times 1.111 = 666.7 \approx 667$ Hz.

9. **B** — The fundamental frequency for a string fixed at both ends is $f_1 = v/(2L)$. (*From formula sheet*) $f_1 = 320/(2 \times 0.80) = 320/1.60 = 200$ Hz.

10. **B** — The beat frequency is $f_{\text{beat}} = |f_1 - f_2| = |440 - 446| = 6$ beats per second. (*Recalled — not on formula sheet*)

Section B: Quantum Physics — Answers

1. **B** — $\langle\alpha|\beta\rangle$ is a scalar (inner product). $|\alpha\rangle$ is a ket vector. $\langle\beta|$ is a bra (dual vector). $|\alpha\rangle\langle\beta|$ is an outer product, which is a **linear operator** that maps kets to kets.
2. **C** — Normalisation requires $\langle\psi|\psi\rangle = 1$. We have $\langle\psi|\psi\rangle = |\alpha|^2(1^2 + 2^2) = 5\alpha^2 = 1$, so $\alpha = 1/\sqrt{5}$.
3. **A** — $\langle\phi|\psi\rangle = \left(\frac{1}{\sqrt{2}}\right)^* \left(\frac{1}{\sqrt{2}}\right) + \left(\frac{i}{\sqrt{2}}\right)^* \left(\frac{-i}{\sqrt{2}}\right) = \frac{1}{2} + \frac{(-i)(-i)}{2} = \frac{1}{2} + \frac{i^2}{2} = \frac{1}{2} + \frac{-1}{2} = 0$. The states are orthogonal.
4. **C** — For an operator to be Hermitian, $\hat{O}^\dagger = \hat{O}$. We have $(|1\rangle\langle 2|)^\dagger = |2\rangle\langle 1|$ and $(i|1\rangle\langle 2|)^\dagger = -i|2\rangle\langle 1|$. So: $\hat{A}^\dagger = -i|2\rangle\langle 1| - i|1\rangle\langle 2| \neq \hat{A}$. $\hat{B}^\dagger = |2\rangle\langle 1| - |1\rangle\langle 2| = -\hat{B} \neq \hat{B}$. $\hat{C}^\dagger = |2\rangle\langle 1| + |1\rangle\langle 2| = \hat{C}$ — Hermitian. $\hat{D}^\dagger = -i|2\rangle\langle 1| + i|1\rangle\langle 2| = -\hat{D} \neq \hat{D}$.
5. **B** — By the Born rule (*Recalled — not on formula sheet*), $P(E_1) = |\langle E_1|\psi\rangle|^2 = (3/5)^2 = 9/25$.
6. **C** — From the formula sheet, $|\uparrow_z\rangle = \frac{1}{\sqrt{2}}|\uparrow_x\rangle + \frac{1}{\sqrt{2}}|\downarrow_x\rangle$. (*From formula sheet*) Therefore $P(S_x = +\hbar/2) = |\langle\uparrow_x|\uparrow_z\rangle|^2 = (1/\sqrt{2})^2 = 1/2$.
7. **C** — Time evolution multiplies each energy eigenstate coefficient by a phase factor $e^{-iE_nt/\hbar}$, which does not change the modulus of the coefficient. (*From formula sheet*) Therefore the probabilities of measuring each energy eigenvalue $P(E_n) = |c_n|^2$ are conserved. The other quantities are generally not conserved unless the corresponding operator commutes with H .
8. **C** — Option A is a product state $|\uparrow_z\rangle_1 |\downarrow_z\rangle_2$. Option B can be written as $\frac{1}{\sqrt{2}}(|\uparrow_z\rangle_1 |\uparrow_z\rangle_2 + |\downarrow_z\rangle_1 |\uparrow_z\rangle_2)$, which factorises as $\frac{1}{\sqrt{2}}(|\uparrow_z\rangle_1 + |\downarrow_z\rangle_1) \otimes |\uparrow_z\rangle_2$ — a product state. Option D is a product state. Option C, $\frac{1}{\sqrt{2}}(|\uparrow_z\rangle_1 |\downarrow_z\rangle_2 - |\downarrow_z\rangle_1 |\uparrow_z\rangle_2)$, cannot be factorised into a product of single-particle states, so it is entangled.
9. **B** — Fermions have antisymmetric states under particle exchange. (*Recalled — not on formula sheet*) The SWAP operator applied to a two-fermion state gives $\text{SWAP}|\Psi\rangle = -|\Psi\rangle$.
10. **C** — The neutron is a baryon with quark content *udd*. (*Recalled — not on formula sheet*) Its charge is $+2/3 + (-1/3) + (-1/3) = 0$, consistent with the neutron being electrically neutral.

Section C: Long Answer Solutions

Question 1.

[15 marks]

Projectile Motion and Energy Conservation

(a) Maximum height 4

The vertical component of the initial velocity is:

$$v_{0y} = v_0 \sin \theta = 25 \sin 53^\circ = 25 \times 0.7986 = 19.97 \text{ m/s}$$

[1]

At maximum height, the vertical velocity is zero. Using $v^2 = u^2 + 2as$ (*From formula sheet*) with $v = 0$, $u = v_{0y}$, $a = -g$:

$$0 = v_{0y}^2 - 2gH$$

[1]

Solving for H :

$$H = \frac{v_{0y}^2}{2g} = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{625 \times 0.6378}{19.62} = \frac{398.6}{19.62}$$

[1]

$$H = 20.3 \text{ m}$$

[1]

(b) Range 3

Using the range formula (derived from SUVAT, with launch and landing at the same height):

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

[1]

Substituting values, with $\sin 2\theta = \sin 106^\circ = 0.9613$:

$$R = \frac{625 \times 0.9613}{9.81} = \frac{600.8}{9.81}$$

[1]

$$R = 61.2 \text{ m}$$

[1]

(c) Maximum compression of the spring 4

The ball is dropped from rest at height $H = 20.3 \text{ m}$. By conservation of energy (*Recalled — not on formula sheet*), the gravitational potential energy at the top equals the elastic potential energy stored in the spring at maximum compression

(taking the ground as the reference level, and noting that at maximum compression the ball is momentarily at rest):

$$mgH = \frac{1}{2}kx^2$$

[1]

(Here we neglect the small additional distance x the ball descends while compressing the spring — or equivalently, we note the problem says the ball falls *onto* the spring on the ground, and for this setup the height from which it falls is H .)

[1]

Solving for x :

$$x^2 = \frac{2mgH}{k} = \frac{2 \times 0.2 \times 9.81 \times 20.3}{500} = \frac{79.66}{500} = 0.1593 \text{ m}^2$$

[1]

$$x = \sqrt{0.1593} = 0.399 \approx 0.40 \text{ m}$$

[1]

(d) **Speed of marble at impact 4**

The energy stored in the compressed spring is:

$$E_{\text{spring}} = \frac{1}{2}kx^2 = \frac{1}{2} \times 500 \times 0.1593 = 39.8 \text{ J}$$

[1]

This energy is converted to kinetic energy of the marble (mass $m_{\text{marble}} = 0.1 \text{ kg}$), giving it a horizontal launch speed:

$$\frac{1}{2}m_{\text{marble}}v_x^2 = E_{\text{spring}} \quad \Rightarrow \quad v_x = \sqrt{\frac{2 \times 39.8}{0.1}} = \sqrt{796} = 28.2 \text{ m/s}$$

[1]

The marble is launched horizontally from the top of a 20 m cliff. The vertical speed acquired in falling $h = 20 \text{ m}$ from rest is:

$$v_y = \sqrt{2gh} = \sqrt{2 \times 9.81 \times 20} = \sqrt{392.4} = 19.8 \text{ m/s}$$

[1]

The speed at impact is the magnitude of the total velocity:

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{28.2^2 + 19.8^2} = \sqrt{795.2 + 392.0} = \sqrt{1187}$$

$$v = 34.5 \text{ m/s}$$

[1]

Question 2.

[15 marks]

Thermodynamic Processes**(a) Work done in isothermal expansion 3**For a reversible isothermal expansion of an ideal gas: (*From formula sheet*)

$$W = nRT \ln\left(\frac{V_2}{V_1}\right)$$

[1]

Substituting:

$$W = 2 \times 8.314 \times 300 \times \ln\left(\frac{0.15}{0.05}\right) = 4988.4 \times \ln 3 = 4988.4 \times 1.0986$$

[1]

$$W = 5480 \text{ J} \approx 5.48 \text{ kJ}$$

[1]

(b) Heat absorbed during isothermal expansion 3For an isothermal process involving an ideal gas, the internal energy depends only on temperature, so $\Delta U = 0$. [1]By the first law of thermodynamics (*Recalled — not on formula sheet*): $\Delta U = Q - W$, so:

$$0 = Q - W \quad \Rightarrow \quad Q = W$$

[1]

$$Q = 5480 \text{ J} \approx 5.48 \text{ kJ}$$

[1]

(c) Final temperature after adiabatic expansion 4For a reversible adiabatic process: (*From formula sheet*)

$$TV^{\gamma-1} = \text{const}$$

[1]

Therefore:

$$T_2 V_2^{\gamma-1} = T_3 V_3^{\gamma-1}$$

[1]

Solving for T_3 , with $\gamma - 1 = 5/3 - 1 = 2/3$:

$$T_3 = T_2 \left(\frac{V_2}{V_3}\right)^{\gamma-1} = 300 \times \left(\frac{0.15}{0.30}\right)^{2/3} = 300 \times (0.5)^{2/3}$$

[1]

$$(0.5)^{2/3} = (2^{-1})^{2/3} = 2^{-2/3} = 1/2^{2/3} = 1/\sqrt[3]{4} = 1/1.587 = 0.630.$$

$$T_3 = 300 \times 0.630 = 189 \text{ K}$$

[1]

(d) **Work done during adiabatic expansion 2**

For an adiabatic process, $Q = 0$, so by the first law: $\Delta U = -W$, i.e., $W = -\Delta U$. [1]

For a monatomic ideal gas, $\Delta U = nC_V(T_3 - T_2) = nC_V\Delta T$:

$$W = -nC_V(T_3 - T_2) = nC_V(T_2 - T_3) = 2 \times \frac{3}{2} \times 8.314 \times (300 - 189)$$

$$W = 2 \times 12.471 \times 111 = 2768.6 \text{ J}$$

$$W \approx 2770 \text{ J} \approx 2.77 \text{ kJ}$$

[1]

(e) **Entropy change of the universe for the isothermal process 3**

The entropy change of the gas during the isothermal expansion: (*From formula sheet*)

$$\Delta S_{\text{gas}} = nR \ln\left(\frac{V_2}{V_1}\right) = 2 \times 8.314 \times \ln 3 = 16.628 \times 1.0986 = 18.27 \text{ J/K}$$

[1]

The gas absorbs heat $Q = 5480 \text{ J}$ from a thermal reservoir at temperature $T = 300 \text{ K}$, so the entropy change of the reservoir is:

$$\Delta S_{\text{reservoir}} = -\frac{Q}{T} = -\frac{5480}{300} = -18.27 \text{ J/K}$$

[1]

The total entropy change of the universe is:

$$\Delta S_{\text{universe}} = \Delta S_{\text{gas}} + \Delta S_{\text{reservoir}} = 18.27 + (-18.27) = 0$$

$\Delta S_{\text{universe}} = 0$. This is consistent with the process being reversible: reversible processes produce no net entropy in the universe, in accordance with the second law of thermodynamics.

[1]

Question 3.

[15 marks]

Damped and Forced Oscillations**(a) Equation of motion and definition of γ 2**

The equation of motion for a damped harmonic oscillator is: (*Recalled — not on formula sheet*)

$$m\ddot{x} + b\dot{x} + kx = 0$$

or equivalently:

$$\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$$

[1]

where the damping parameter is defined as:

$$\gamma = \frac{b}{m}$$

[1]

(b) Calculate ω_0 , γ , and ω_d 3

The natural angular frequency: (*From formula sheet*)

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.5}} = \sqrt{400} = 20 \text{ rad/s}$$

[1]

The damping parameter: (*Recalled — not on formula sheet*)

$$\gamma = \frac{b}{m} = \frac{2}{0.5} = 4 \text{ s}^{-1}$$

[1]

The damped angular frequency: (*From formula sheet*)

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2/4} = \sqrt{400 - 16/4} = \sqrt{400 - 4} = \sqrt{396}$$

$$\omega_0 = 20 \text{ rad/s}, \gamma = 4 \text{ s}^{-1}, \omega_d = \sqrt{396} \approx 19.9 \text{ rad/s}$$

[1]

(c) Quality factor and damping regime 2

The quality factor: (*From formula sheet*)

$$Q = \frac{\omega_0}{\gamma} = \frac{20}{4} = 5$$

[1]

To determine the damping regime, compare γ with $2\omega_0$: (*Recalled — not on formula sheet*)

$$\gamma = 4 \quad \text{vs} \quad 2\omega_0 = 40$$

Since $\gamma < 2\omega_0$, the system is **underdamped**. (Equivalently, $\omega_d^2 = \omega_0^2 - \gamma^2/4 = 396 > 0$, confirming oscillatory behaviour.)

$Q = 5$. The system is underdamped.

[1]

(d) **Steady-state amplitude at $\omega = 19$ rad/s** 4

The steady-state amplitude for a driven damped oscillator is: (*From formula sheet*)

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2}}$$

[1]

Compute the required quantities:

$$\frac{F_0}{m} = \frac{5}{0.5} = 10 \text{ m/s}^2$$

[1]

$$\omega_0^2 - \omega^2 = 400 - 361 = 39$$

$$\gamma^2\omega^2 = 16 \times 361 = 5776$$

$$(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2 = 39^2 + 5776 = 1521 + 5776 = 7297$$

[1]

$$A(19) = \frac{10}{\sqrt{7297}} = \frac{10}{85.4}$$

$$A(19) = 0.117 \text{ m} \approx 11.7 \text{ cm}$$

[1]

(e) **Resonance frequency and maximum amplitude** 4

The amplitude resonance occurs at the driving frequency that maximises $A(\omega)$. To find this, minimise the denominator $(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2$ with respect to ω .

Let $u = \omega^2$. Then we minimise $f(u) = (\omega_0^2 - u)^2 + \gamma^2u$:

$$\frac{df}{du} = -2(\omega_0^2 - u) + \gamma^2 = 0 \quad \Rightarrow \quad u = \omega_0^2 - \frac{\gamma^2}{2}$$

[1]

Therefore:

$$\omega_{\text{res}} = \sqrt{\omega_0^2 - \frac{\gamma^2}{2}} = \sqrt{400 - 8} = \sqrt{392}$$

$$\omega_{\text{res}} = \sqrt{392} \approx 19.8 \text{ rad/s}$$

[1]

At this frequency, the denominator is:

$$\begin{aligned}(\omega_0^2 - \omega_{\text{res}}^2)^2 + \gamma^2 \omega_{\text{res}}^2 &= \left(\frac{\gamma^2}{2}\right)^2 + \gamma^2 \left(\omega_0^2 - \frac{\gamma^2}{2}\right) = \frac{\gamma^4}{4} + \gamma^2 \omega_0^2 - \frac{\gamma^4}{2} \\&= \gamma^2 \omega_0^2 - \frac{\gamma^4}{4} = \gamma^2 \left(\omega_0^2 - \frac{\gamma^2}{4}\right) = \gamma^2 \omega_d^2\end{aligned}$$

[1]

So the maximum amplitude is:

$$A_{\text{max}} = \frac{F_0/m}{\gamma \omega_d} = \frac{10}{4 \times \sqrt{396}} = \frac{10}{4 \times 19.9} = \frac{10}{79.6}$$

$$A_{\text{max}} = 0.126 \text{ m} \approx 12.6 \text{ cm}$$

[1]

Question 4.

[15 marks]

Spin-1/2 and Measurement**(a) Normalisation and S_z measurement 3**Check normalisation: (*Recalled — not on formula sheet*)

$$\langle\psi|\psi\rangle = \left|\frac{1}{\sqrt{3}}\right|^2 + \left|\sqrt{\frac{2}{3}}\right|^2 = \frac{1}{3} + \frac{2}{3} = 1 \quad \checkmark$$

[1]

The possible outcomes of an S_z measurement are the eigenvalues of S_z :

[1]

- $S_z = +\hbar/2$ with probability $P(\uparrow_z) = |1/\sqrt{3}|^2 = 1/3$
- $S_z = -\hbar/2$ with probability $P(\downarrow_z) = |\sqrt{2/3}|^2 = 2/3$

The state is normalised. Outcomes: $+\hbar/2$ with probability $1/3$; $-\hbar/2$ with probability $2/3$.

[1]

(b) Express $|\psi\rangle$ in the S_x eigenbasis 4From the formula sheet: (*From formula sheet*)

$$|\uparrow_z\rangle = \frac{1}{\sqrt{2}}(|\uparrow_x\rangle + |\downarrow_x\rangle), \quad |\downarrow_z\rangle = \frac{1}{\sqrt{2}}(|\uparrow_x\rangle - |\downarrow_x\rangle)$$

[1]

Substituting into $|\psi\rangle$:

$$\begin{aligned} |\psi\rangle &= \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}}(|\uparrow_x\rangle + |\downarrow_x\rangle) + \sqrt{\frac{2}{3}} \cdot \frac{1}{\sqrt{2}}(|\uparrow_x\rangle - |\downarrow_x\rangle) \\ &= \frac{1}{\sqrt{6}}(|\uparrow_x\rangle + |\downarrow_x\rangle) + \frac{\sqrt{2}}{\sqrt{6}}(|\uparrow_x\rangle - |\downarrow_x\rangle) \end{aligned}$$

[1]

$$= \frac{1 + \sqrt{2}}{\sqrt{6}} |\uparrow_x\rangle + \frac{1 - \sqrt{2}}{\sqrt{6}} |\downarrow_x\rangle$$

[1]

The probabilities of each S_x outcome are:

$$P(S_x = +\hbar/2) = \left(\frac{1 + \sqrt{2}}{\sqrt{6}}\right)^2 = \frac{(1 + \sqrt{2})^2}{6} = \frac{1 + 2\sqrt{2} + 2}{6} = \frac{3 + 2\sqrt{2}}{6} \approx 0.971$$

$$P(S_x = -\hbar/2) = \left(\frac{1 - \sqrt{2}}{\sqrt{6}}\right)^2 = \frac{(1 - \sqrt{2})^2}{6} = \frac{1 - 2\sqrt{2} + 2}{6} = \frac{3 - 2\sqrt{2}}{6} \approx 0.029$$

Check: $(3 + 2\sqrt{2})/6 + (3 - 2\sqrt{2})/6 = 6/6 = 1 \checkmark$

$$P(S_x = +\hbar/2) = (3 + 2\sqrt{2})/6 \approx 0.971; \quad P(S_x = -\hbar/2) = (3 - 2\sqrt{2})/6 \approx 0.029.$$

[1]

(c) **Post-measurement S_z probabilities** 4

After measuring $S_x = +\hbar/2$, the state collapses to:

[1]

$$|\uparrow_x\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$$

(From formula sheet)

[1]

If S_z is now measured on $|\uparrow_x\rangle$:

[1]

$$P(S_z = +\hbar/2) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

$$P(S_z = -\hbar/2) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

$$P(S_z = +\hbar/2) = 1/2; \quad P(S_z = -\hbar/2) = 1/2.$$

[1]

(d) **Eigenvalues, eigenvectors, and Hermiticity of $\hat{A}$** 4

In the $\{|\uparrow_z\rangle, |\downarrow_z\rangle\}$ basis, the matrix representation of $\hat{A}$ is:

$$\hat{A} = \hbar \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

[1]

(Note: this is $\hbar\sigma_x$ where σ_x is the Pauli- x matrix from the formula sheet.)

Finding eigenvalues from $\det(\hat{A} - \lambda I) = 0$:

$$\det \begin{pmatrix} -\lambda & \hbar \\ \hbar & -\lambda \end{pmatrix} = \lambda^2 - \hbar^2 = 0 \quad \Rightarrow \quad \lambda = \pm\hbar$$

[1]

For $\lambda = +\hbar$: $\hat{A}|v\rangle = \hbar|v\rangle$ gives $|v_+\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle) = |\uparrow_x\rangle$.

For $\lambda = -\hbar$: $\hat{A}|v\rangle = -\hbar|v\rangle$ gives $|v_-\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle - |\downarrow_z\rangle) = |\downarrow_x\rangle$.

[1]

To check Hermiticity: $\hat{A}^\dagger = \hbar(|\downarrow_z\rangle\langle\uparrow_z| + |\uparrow_z\rangle\langle\downarrow_z|) = \hat{A}$.

Alternatively, the matrix $\hbar \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is real and symmetric, hence Hermitian.

Eigenvalues: $\lambda = +\hbar$ and $\lambda = -\hbar$. Eigenvectors: $|v_+\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle + |\downarrow_z\rangle)$ and $|v_-\rangle = \frac{1}{\sqrt{2}}(|\uparrow_z\rangle - |\downarrow_z\rangle)$. $\hat{A}$ is Hermitian since $\hat{A}^\dagger = \hat{A}$.

[1]